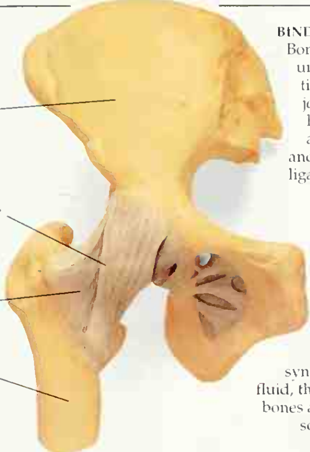
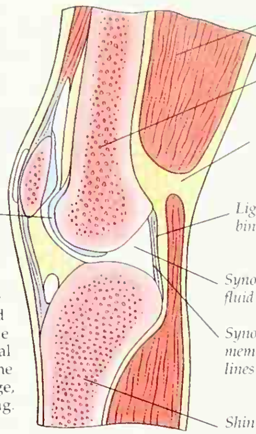


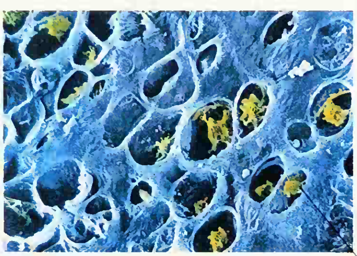
BINDING THE BONES
Bones do not move about freely in a joint, unrestrained. Tough straps of strong elastic tissue known as ligaments surround the joint, and bind the bones. The ligaments hold the bones securely against each other, and prevent them moving excessively and getting dislocated, or out of joint. These ligaments bind the front of the right hip joint.



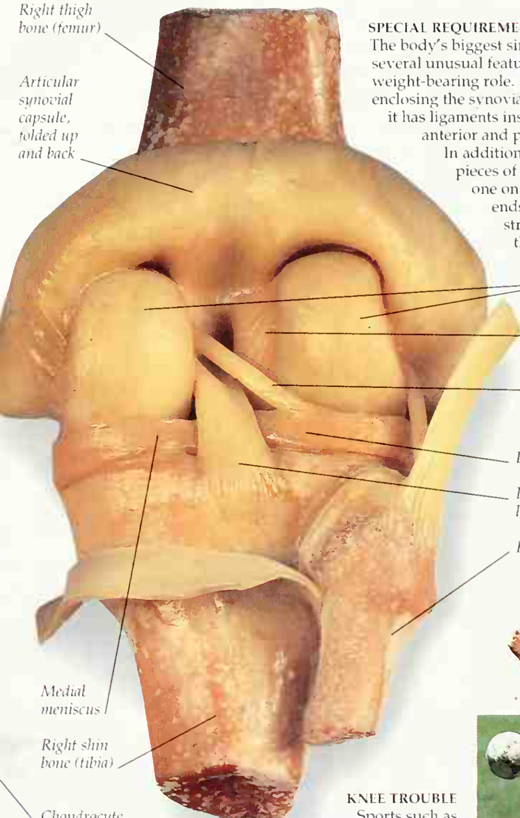
INSIDE A SYNOVIAL JOINT
This view into the knee shows the main parts of a movable joint. Inside the ligaments is a stretchy bag called the synovial capsule. The bag's lining is the synovial membrane. This makes slippery synovial fluid, the "oil" that lubricates the joint. The ends of the bones are covered in friction-reducing shiny cartilage, so the whole joint is strong and smooth running.

Cartilage

There are several types of cartilage, or gristle, in the body. The main type is hyaline cartilage. It is a pearly light blue and makes up structural parts such as the nose, voice box, windpipe, and lung airways (pp. 24-25). Inside a joint, a specialized hyaline cartilage, called articular hyaline cartilage, covers the ends of bones where they press and rub against each other. It is glossy to reduce friction and slightly spongy to absorb jolts. Another type is white fibrocartilage, which is found in disks in between the vertebrae (p. 14) and in parts of certain joints, such as the sockets in the shoulder and hip bones, to withstand the immense wear. A third type, found in parts of the ears and the larynx, is springy and elastic. It is called yellow fibrocartilage.



CARTILAGE CELLS
Cartilage-making cells are called chondrocytes. They live entombed in the matrix that they make around themselves. This is composed of fibers of the proteins collagen and elastin, woven into a stiff jelly with water, dissolved carbohydrates, such as starches, and body minerals. Cartilage has a limited blood supply, and nutrients seep in and wastes seep out to blood vessels around the edges.



SPECIAL REQUIREMENTS
The body's biggest single joint, the knee, has several unusual features related to its great weight-bearing role. Besides the usual ligaments enclosing the synovial capsule outside the joint, it has ligaments inside the joint. They are the anterior and posterior cruciate ligaments. In addition, two extra crescent-shaped pieces of cartilage float in the joint, one on each side between the bone ends. These menisci add more strength and stability, helping the knee to "lock" straight.

KNEE TROUBLE
Sports such as soccer involve swiveling on the legs and high kicks, so players like Diego Maradona may tear or displace the menisci cartilages. This is often referred to as knee cartilage trouble. In some cases one or both menisci are removed completely, as the knee can function reasonably without them.